

What is OS/COMET®

OS/COMET® is a COTS software framework for spacecraft telemetry, tracking, and control (TT&C). It delivers a turnkey command-and-control (C2) solution that supports:

- Satellite command and control
- Payload/sensor monitoring and control
- Ground equipment monitoring and control
- Event response and schedule-based automation
- Automated operations (ground and space) with human-in-the-loop options
- Automated integration and test (I&T) systems for rapid verification and validation

Designed for flexibility, OS/COMET® enables seamless integration with existing systems and applications or the rapid development of new ones to support any spacecraft, ground system, or mission type.

Whether modernizing a legacy ground system, standing up a new mission, or scaling to multi-payload operations, OS/COMET® provides the proven foundation you need built for mission assurance, designed for flexibility, and ready for tomorrow. Peraton is ready to help you accelerate your mission timeline.

WHY CHOOSE OS/COMET®

- **Proven at Scale**
Operationally deployed for decades across civil, defense, and commercial missions.
- **Modular & Extensible**
A framework that adapts to unique mission needs, from single spacecraft to complex constellations.
- **Interoperable**
Works with legacy and modern interfaces and adheres to industry standards, reducing integration friction.
- **Operator Centric**
Streamlined workflows, role-based access, and intuitive UI reduce training time and operator workload.
- **Automation Where It Matters**
Event-driven and schedule-based automation minimizes manual tasks while preserving human oversight when required.
- **Future Ready**
Built to support next-generation constellations, hosted payloads, and evolving mission profiles.

TECHNICAL CAPABILITIES

- **Command & Control**
Robust command sequencing, real-time telemetry visualization, and safe command execution guardrails to prevent operator error.
- **Payload & Sensor Ops**
Configurable dashboards and rules enable precise payload control and health monitoring, improving on-orbit utilization.
- **Ground Segment Control**
Integrated ground equipment monitoring/control to coordinate networks, antennas, and RF chains for end-to-end situational awareness.
- **Event Response & Automation**
Trigger actions on thresholds and anomalies; schedule routine contacts, calibrations, and maintenance to reduce staffing burdens.
- **Human-in-the-Loop Ops**
Insert approvals at critical steps, enforce permissions, and log actions for auditability across mission operations.
- **Integration & Test**
Automated test execution, data capture, and regression reporting accelerates I&T cycles and reduce risk during transitions to operations.

DEPLOYMENT FLEXIBILITY

OS/COMET® has been deployed in operational environments on bare metal, virtualized hardware, and cloud infrastructure, supporting hybrid architectures where needed. Typical patterns include:

- On-premises deployments for highly controlled environments
- Virtualized deployments for resource efficiency and isolation
- Cloud deployments on leading platforms, with options to leverage government clouds and secure enclaves
- High availability & disaster recovery architectures to maintain mission continuity

SECURITY & COMPLIANCE

Operational environments for OS/COMET® deployments align to NIST SP 800-53 and DFARS security requirements. The platform is designed to support:

- Role-based access control, auditing, and least-privilege operations
- Encryption in transit and at rest according to mission policy
- Network segmentation and hardened configurations to meet mission risk postures
- Deployment within environments that follow RMF, STIG-hardening, and customer-specific ATO processes

TRUSTED ACROSS CRITICAL MISSIONS

Trusted by U.S. government and commercial operators, Peraton supports missions including:

- NOAA's GOES series of weather satellites
- Multiple U.S. Space Force missions, including GPS
- Human spaceflight systems such as the Orion spacecraft

With OS/COMET®, we currently enable operations for 100+ satellites and 250 hosted payloads worldwide. Commercial customers include Vantor, Telesat, and hosted payload Aireon and its auxiliary components to include Iridium NEXT, Road Runner, SeaSmart, and Reach.

WHAT CAN OS/COMET® DO FOR YOU

- Reduced operator workload through automation and efficient workflows
- Faster integration & test cycles and smoother transitions to operations
- Improved mission resilience via standardized processes and observability
- Lower lifecycle costs through COTS maintainability and extensibility

BUILT FOR WHAT'S NEXT

As space operations evolve, OS/COMET® continues to advance to support:

- Mega constellations and multi-mission operations centers
- Autonomous operations with policy driven safeguards
- Advanced analytics integrations for trend, anomaly, and performance insights
- Emerging mission domains, including cislunar and deep-space operations